Financial Variance Dashboard: Build Guide

Read this first: there is no budget in this dataset

The Microsoft Financial Sample has actuals only (700 rows, Sep 2013 to Dec 2014). To compare actual vs budget, this guide builds the budget from **two editable planning assumptions**: a planned discount rate of 5% of Gross Sales, and a target COGS ratio of 82% of Net Sales. Both are stated as assumptions, not data. Say so in your report. If your course gives you a real budget file, replace the budget measures in section 3 with it.

Cost centres: the dataset has no cost-centre field, so Product (Amarilla, Carretera, Montana, Paseo, VTT, Velo) is used as the manufacturing cost centre, since COGS is recorded per product.

1. Load the data

Get Data > Text/CSV, load all 6 files from financial_star.zip: Fact_Financials, Dim_Date, Dim_Product, Dim_Segment, Dim_Country, Dim_DiscountBand. Set types: Date = Date; Units Sold, prices, Gross Sales, Discounts, Net Sales, COGS, Profit = Decimal. Note: *Net Sales* is the original dataset column called "Sales", renamed to avoid confusion. Discount Band "None" is a real value, not a null.

2. Model

One side	Many side (Fact_Financials)
Dim_Date[Date]	[Date] (mark Dim_Date as date table)
Dim_Product[Product]	[Product]
Dim_Segment[Segment]	[Segment]
Dim_Country[Country]	[Country]
Dim_DiscountBand[Discount Band]	[Discount Band] (sort by Discount Band Order)

3. DAX measures

Income statement (Gross Sales to Profit)

```
Gross Sales = SUM ( Fact_Financials[Gross Sales] )
Discounts = SUM ( Fact_Financials[Discounts] )
Net Sales = SUM ( Fact_Financials[Net Sales] )
COGS = SUM ( Fact_Financials[COGS] )
Gross Profit = [Net Sales] - [COGS]

Gross Profit % = DIVIDE ( [Gross Profit], [Net Sales], 0 )
COGS Ratio = DIVIDE ( [COGS], [Net Sales], 0 )
Discount Rate % = DIVIDE ( [Discounts], [Gross Sales], 0 )
Segment Margin = DIVIDE ( [Gross Profit], [Net Sales], 0 )   -- use with Segment on rows/axis
Segment Profit Share = DIVIDE ( [Gross Profit], CALCULATE ( [Gross Profit], ALL ( Dim_Segment ) ), 0 )
```

Budget and variance (budget = assumptions)

```
Budget Discount % = 0.05
Budget COGS Ratio = 0.82

Budget Gross Sales = [Gross Sales]
Budget Discounts = [Budget Gross Sales] * [Budget Discount %]
Budget Net Sales = [Budget Gross Sales] - [Budget Discounts]
Budget COGS = [Budget Net Sales] * [Budget COGS Ratio]
Budget Profit = [Budget Net Sales] - [Budget COGS]

Net Sales Variance = [Net Sales] - [Budget Net Sales]
Net Sales Variance % = DIVIDE ( [Net Sales Variance], [Budget Net Sales] )
Profit Variance = [Gross Profit] - [Budget Profit]
Profit Variance % = DIVIDE ( [Profit Variance], [Budget Profit] )
```

4. Waterfall chart (revenue erosion)

Home > Enter Data: create a table **Waterfall Items** with one column Item and rows: Gross Sales, Discounts, COGS. Add the measure below, then build a native **Waterfall chart**: Category = Waterfall Items[Item], Y = [Waterfall Value].

Power BI adds the final "Total" bar automatically, and it equals Gross Profit.

```
Waterfall Value =
SWITCH ( SELECTEDVALUE ( 'Waterfall Items'[Item] ),
    "Gross Sales", [Gross Sales],
    "Discounts",  -[Discounts],
    "COGS",       -[COGS] )
```

Sort Item manually (Gross Sales, Discounts, COGS) using a Sort column in the table.

5. Scenario forecasting sliders

Modeling > New parameter > Numeric range, tick "Add slicer". Create three: **Volume Change %** (-20% to +20%, step 1%, default 0), **Scenario Discount %** (0% to 20%, step 0.5%, default 7%), **Scenario COGS Ratio** (70% to 95%, step 0.5%, default 86%). Each creates a measure named like [Volume Change % Value]. Then:

```
Scenario Gross Sales = [Gross Sales] * ( 1 + [Volume Change % Value] )
Scenario Net Sales = [Scenario Gross Sales] * ( 1 - [Scenario Discount % Value] )
Scenario Profit = [Scenario Net Sales] * ( 1 - [Scenario COGS Ratio Value] )
Scenario vs Actual Profit = [Scenario Profit] - [Gross Profit]
Scenario vs Budget Profit = [Scenario Profit] - [Budget Profit]
```

Put the three sliders and cards for Scenario Net Sales, Scenario Profit and Scenario vs Budget Profit on the page.

6. Page layout

Visual	Setup
KPI cards	Gross Sales, Net Sales, Gross Profit, Gross Profit %, COGS Ratio
Waterfall	Section 4
Variance by Segment	Clustered bar: Axis = Segment, Values = Net Sales and Budget Net Sales; a second visual for Profit Variance
Variance by Country	Map or bar: Country, Profit Variance (red/green conditional colour)
Cost centre matrix	Matrix: Rows = Product, Columns = Segment, Values = Segment Margin. Conditional formatting red to green.
Discount analysis	Column chart: Discount Band vs Gross Profit % (High band is the weakest to check)
Slicers	Year, Country, Segment, plus the 3 scenario sliders

7. Numbers to check against (no filters)

Measure	Expected
Gross Sales	$127.93M
Discounts	$9.21M (7.2% of Gross Sales)
Net Sales	$118.73M
COGS	$101.83M (COGS Ratio 85.8%)
Gross Profit	$16.89M (Gross Profit % 14.2%)
Budget Net Sales / Budget Profit	$121.54M / $21.88M
Net Sales Variance / Profit Variance	-$2.81M / -$4.98M

8. What the data says (for your insights section)

Enterprise is the problem segment: $19.6M net sales but a -$0.61M profit (-3.1% margin), about -$4.2M versus budget profit. Small Business is also under budget (about -$3.7M). Government is the main profit engine at $11.4M (21.7% margin). Channel Partners has a 73% margin but only $1.8M in sales. 58 of 700 rows are loss-making. Every product misses the profit budget, so the gap comes from the COGS ratio (85.8% vs the 82% target) more than from volume.

Caveat: results versus budget depend on the two assumptions above, so state them next to every variance figure.